Reproducing and Critically Reassessing Spectral Shrunk Clustering: Implementation Corrections and a Mutual k-NN Adaptation

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Abstract

This course report reimplements Spectral Shrunk Clustering (SSC) of Chang, Nie, Ma, Yang, and Zhou from the published equations, without original author code. SSC first computes a normalized-cut embedding and then shrinks that embedding with a convex $\ell_{2,1}$ fidelity term and a graph fusion penalty. We document two typesetting inconsistencies—Equation (10) writes an embedding-dimension diagonal that cannot multiply in the $n \times n$ trace of Equation (9), and Algorithm 1 line 10 replaces $\widetilde{L}$ by $\widetilde{W}$ and F by X—and we implement the stationary point printed as Equation (13). That fixed point matches Equation (8) with graph coefficient $\gamma/2$, which we treat as a correction rather than a contribution. Four of the eight paper datasets were obtained with checksums (JAFFE and USPS verified; ORL and YaleB approximate variants); AR, UMIST, BinAlpha, and MSRA50 were not obtained. SPLS is not implemented. On the obtained data, SSC did *not* recover the paper's large ORL gain: our Track A SSC accuracy is 53.9 ± 3.0 ($n = 50$) versus the published 76.0 ± 1.6, and is indistinguishable from our spectral clustering baseline (54.2 ± 3.1). YaleB SSC used 10 trials, not 50; USPS SSC used 1 trial, not 50. Replacing the shrinkage graph by mutual kNN is mixed or negligible and is not a win.

1 Introduction

Clustering partitions points so that nearby samples share a label. On images and other high-dimensional observations, Euclidean distances in the ambient space are often uninformative, and methods that first reduce dimension become attractive. Spectral clustering [Shi and Malik, 2000, von Luxburg, 2007] builds a similarity graph, forms a Laplacian, and uses a few eigenvectors as coordinates in which ordinary K-means can succeed. Those coordinates are a discrete analogue of a manifold embedding: they try to preserve neighbourhood structure rather than global variance, which is the same distinction COMP3314 draws between principal component analysis (PCA) and Laplacian-style geometry.

Most spectral pipelines, however, construct the graph in the original feature space and only then embed. Chang et al. argue that neighbourhoods in that space need not match neighbourhoods after the embedding, so a Laplacian built from the original x_i can be the wrong regularizer for the reduced

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points f_i. Spectral Shrunk Clustering (SSC) therefore computes a normalized-cut embedding F, rebuilds a k-nearest-neighbour (kNN) graph on F, and solves a convex shrinkage problem whose solution G is clustered by K-means [Chang et al., 2015]. The shrinkage combines an $\ell_{2,1}$ fidelity term, intended to be robust to row-wise outliers [Nie et al., 2010], with a weighted fusion penalty in the spirit of pattern shrinking [Hou et al., 2013] and convex clustering [Chi and Lange, 2015].

This report has four aims, and we keep them separate. First, we reimplement SSC from the paper's equations, recording every unspecified quantity as an assumption and implementing Equation (13) rather than Algorithm 1 line 10. Second, we obtain the paper's datasets only from traceable files, which currently yields four collections and four unavailable names. Third, we compare published claims with our protocol rather than copying Table 1–2 of Chang et al. into a "reproduction" column. Fourth, we adapt the shrinkage affinity to a mutual kNN graph on F [Brito et al., 1997, von Luxburg, 2007], freeze those nonnegative weights before iteratively reweighted least squares (IRLS), and report a mixed/negligible empirical difference rather than a gain. No new convexity theorem is claimed.

2 Background and related work

Lloyd K-means [Lloyd, 1982] alternates assignment to nearest centres with centre updates. It is the final discretization in SSC and in the spectral baseline; COMP3314 treats it as the canonical partitioning method against which linear projections are compared. PCA followed by K-means keeps directions of largest variance. Linear discriminant analysis (LDA) followed by K-means, in the unsupervised variant of Ding and Li [2007], replaces those directions by discriminants computed from the *current* cluster labels, never from ground truth. Both constructions appear in Chang et al.; they connect directly to the PCA and LDA material in COMP3314, with the caveat that classical LDA has rank at most $c - 1$.

Spectral clustering replaces the linear subspace with eigenvectors of a graph Laplacian. Shi and Malik [2000] introduce the normalized cut. von Luxburg [2007] separate the unnormalized Laplacian $L = D - A$ from the symmetric normalized Laplacian $L_n = I - D^{-1/2}AD^{-1/2}$ and warn that the choice of graph (union kNN, mutual kNN, or ε-ball) already changes the cut. Ng et al. [2001] add a row-normalization step that SSC does not write. Self-tuning spectral clustering [Zelnik-Manor and Perona, 2004], which Chang et al. cite, replaces a global bandwidth by local scales; SSC still uses one δ per graph. Local learning clustering [Wu and Schölkopf, 2006] changes the Laplacian itself, predicting each label from neighbours. Adaptive-neighbour methods [Nie et al., 2014] learn the graph jointly with the partition. SSC does neither: after F is fixed, it freezes a Gaussian kNN weight W and shrinks F.

Pattern shrinking [Hou et al., 2013] pulls representations toward one another while staying close to an initial embedding; Chang et al. cite it as SPLS (the source paper's own acronym is SLPS). Convex clustering [Chi and Lange, 2015] uses a sum-of-norms fusion penalty on centroids of the original points, with weights chosen before the path is computed. Robust $\ell_{2,1}$ losses [Nie et al., 2010] downweight entire sample rows. IRLS turns both the $\ell_{2,1}$ term and the unsquared fusion term into quadratic surrogates whose weights depend on the current G. Those COMP3314 ingredients—graphs, eigenmaps, convex surrogates, and unsupervised linear projections—are exactly the pieces SSC assembles.

3 Revisiting spectral shrunk clustering

Notation. Samples are rows of $X \in \mathbb{R}^{n\times d}$. Algorithm 1 of Chang et al. writes $X \in \mathbb{R}^{d\times n}$; Equation (13) never multiplies by X, so the solver does not use that layout. Let c be the number of clusters. The spectral embedding is $F \in \mathbb{R}^{n\times c}$ with rows f_i, and the shrunk matrix is $G \in \mathbb{R}^{n\times c}$ with rows g_i.

Spectral embedding. A Gaussian kNN graph is built on the original features. Chang et al. set $k = 5$ for every spectral method. The affinity is $A_{ij} = \exp(-\|x_i - x_j\|^2/\delta^2)$ when i and j are neighbours and 0 otherwise. The paper does not specify δ; we take δ^2 to be the median of the positive directed kNN squared distances on that graph. Undirected edges are the union of the two directed relations [von Luxburg, 2007]. The symmetric normalized Laplacian is

$$L_n = I - D^{-1/2}AD^{-1/2}. \tag{1}$$

F contains the c eigenvectors for the smallest eigenvalues of L_n, ordered by increasing eigenvalue. Isolated vertices (degree zero) receive $D_{ii}^{-1/2} = 0$, which the paper does not discuss. Eigenvector signs are fixed so that the entry of largest magnitude in each column is positive. This is not PCA.

Shrinkage objective. A second kNN Gaussian graph W is built on the rows of F, with its own median bandwidth. The printed convex problem is

$$\min_G \|G - F\|_{2,1} + \gamma \sum_{ij} W_{ij} \|g_i - g_j\|_2, \tag{2}$$

where $\|H\|_{2,1} = \sum_i \|h_i\|_2$ and h_i is row i of $H = G - F$. The sum runs over ordered pairs; both directed copies of a symmetric edge are counted.

Equation (10) is dimensionally inconsistent. Equation (9) writes $\mathrm{Tr}((G-F)^\top S(G-F))$, which requires $S \in \mathbb{R}^{n \times n}$. Equation (10) nevertheless sets

$$S = \mathrm{diag}\big(1/(2\|h_1\|_2), \ldots, 1/(2\|h_d\|_2)\big)$$

and calls d the embedding dimension. If the h_j are columns, S is $c \times c$ (or $d \times d$ in the original feature dimension), which cannot sit between $(G-F)^\top$ and $(G-F)$. Row-wise $\ell_{2,1}$ as used by Nie et al. [2010] also acts on samples, not on embedding coordinates. We therefore use the $n \times n$ diagonal

$$S_{ii} = \frac{1}{2\max(\|h_i\|_2, \varepsilon)}, \tag{3}$$

with ε only to avoid division by zero. This is a correction of the printed formula, not a new regularizer.

Algorithm 1 line 10 versus Equation (13). With $\widetilde{W}_{ij} = W_{ij}/(2\|g_i - g_j\|_2)$ and combinatorial Laplacian $\widetilde{L} = \widetilde{D} - \widetilde{W}$, the quadratic surrogate is Equation (12). Setting its derivative to zero yields

$$G = (S + \gamma \widetilde{L})^{-1} SF. \tag{4}$$

Algorithm 1 line 10 instead prints $G_{t+1} = (S_t + \gamma \widetilde{W})^{-1} S_t X$. That line replaces $\widetilde{L}$ by $\widetilde{W}$ and F by X, and line 4 computes $\widetilde{W}$ outside the loop even though Equation (11) depends on G. We solve (4) with a sparse symmetric solve and recompute $\widetilde{W}$ and $\widetilde{L}$ from the current G at every iteration. We do not form an explicit inverse.

The implemented stationary point is $\gamma/2$, not printed Equation (8). For symmetric W, $\sum_{ij} \widetilde{W}_{ij} \|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L} G)$. The IRLS majorizer of the graph term in (2) is therefore stationary at $(S + 2\gamma\widetilde{L})G = SF$. Direct differentiation of (2) gives the same factor of four on $\widetilde{L}G$ versus two on $S(G-F)$. Equation (13) uses $\gamma\widetilde{L}$, so its fixed point is the convex problem that replaces the graph coefficient in (2) by $\gamma/2$. A unit-test comparison against an independent convex solver on a six-point chain is recorded in Appendix A. We lock the implementation to printed Equation (13). That choice is a correction of the paper's internal mismatch, not a new method.

Initialization, stopping, and K-means. The paper writes $G_0 \in \mathbb{R}^{n \times c}$ and does not give a matrix. We set $G_0 = F$. Then $H_0 = 0$ and Equation (10) is undefined. Using ε as the first denominator keeps G at F, which is generally not a stationary point of (2). The default therefore uses, only on the first all-zero residual, the median pairwise row distance of the current G as a finite scale, and ε thereafter. The paper gives no stopping rule; we stop on a relative Frobenius step of 10^{-6} or after 100 iterations. Labels come from K-means on G and are never written back into W, F, or G.

4 Reproduction methodology

4.1 Datasets and provenance

Chang et al. name eight collections. We download only files with a URL, a date, and a SHA-256 recorded in each dataset package's `manifest.yaml`. Synthetic arrays are restricted to unit tests

Table 1: Datasets named by Chang et al. Counts n, c, and d are measured on the local file when a file exists, and are left blank when no archive was obtained. Paper claims for unavailable sets are not copied into the count columns.

Dataset	Status	n	c	d	Notes
AR	unavailable	—	—	—	access URLs failed
JAFFE	verified	213	10	676	Zenodo TIFF $\to 26 \times 26$
ORL	approximate	400	40	1024	Olivetti $64 \times 64 \to 32 \times 32$
UMIST	unavailable	—	—	—	WAF / 404; no archive
BinAlpha	unavailable	—	—	—	no 26×30 file
MSRA50	unavailable	—	—	—	no $1799/12$ file
YaleB	approximate	2414	38	1024	Deng Cai 32×32 mat
USPS	verified	9298	10	256	LIBSVM 16×16

and are never renamed to a paper dataset. Table 1 summarizes status; Appendix B lists sources and licences.

ORL is the scikit-learn Olivetti matrix [Samaria and Harter, 1994, Pedregosa et al., 2011]: 400 images, 40 people, stored at 64×64, then bilinearly resized to 32×32. Common ORL releases are 92×112; this is not that array. JAFFE is the Zenodo TIFF archive of 213 images of 10 posers [Lyons et al., 1998, 2025], verified at 256×256 and resized to 26×26 as the paper states; interpolation is unspecified in the paper and is bilinear here. Redistribution of JAFFE is forbidden. YaleB is Deng Cai's `YaleB_32x32.mat` with 2414 images and 38 people [Georghiades et al., 2001, Lee et al., 2005]. Official cropped PGM zips returned HTTP 404 on 2026-10-20, so the resize kernel from 168×192 is unknown. USPS is the LIBSVM pair `usps.bz2` and `usps.t.bz2` [Hull, 1994]: 9298 samples, 16×16, already in that size, with class counts matching the standard split after concatenating train then test.

AR [Martinez and Benavente, 1998] requires a university-email request; the public HTML endpoints returned 503 or 404, and we did not obtain credentials. UMIST / Sheffield [Graham and Allinson, 1998] was blocked by a WAF challenge. BinAlpha is cited without a source; Binary Alphadigits (36 classes) was identified and not substituted, because the paper's 26×30 subset cannot be reconstructed without the parent pool and seed. MSRA50 is cited without a URL; MSRA-B, MSRA10K, and related names are different collections and were not renamed.

Preprocessing is grayscale conversion where needed, bilinear resize when the paper states a size, vectorization, and division by 255 (USPS is left in its stored $[-1, 1]$ scale). There is no augmentation and no neural embedding.

4.2 Baselines

K-means, spectral clustering from Equation (1), PCA+K-means, PCA+spectral, LDA+K-means [Ding and Li, 2007], and local learning clustering [Wu and Schölkopf, 2006] are reimplemented with a common `fit_predict` interface that never reads ground-truth labels. LDA+spectral clustering is marked `APPROXIMATE`: Ding and Li specify a K-means inner step, and Chang et al. do not say whether LDA is fit once or alternated. SPLS [Hou et al., 2013] is `NOT_IMPLEMENTED`; the open abstract does not contain the Section 3 updates, and result cells for SPLS must read "N/A — not reproduced" rather than numbers from another method. LDA methods raise if $n_{\text{components}} > c - 1$, so the paper's "10 to 150" subspace sentence is not feasible for classical LDA on a 10-class set such as USPS.

4.3 Tracks, metrics, and compute

Protocol file `benchmarks/protocols/original_paper.yaml` distinguishes paper text, inferences, and our assumptions.

Track A follows the paper's "report the best results" sentence: $k = 5$; $\gamma \in \{10^{-6}, 10^{-3}, 1, 10^{3}, 10^{6}\}$; subspace dimensions in $\{10, 20, \ldots, 150\}$ clipped by n and d. Model selection uses labels and is therefore not a fair unsupervised tune. Track B uses the same files but no labels for training or tuning:

$\gamma = 1$ is frozen in advance (the paper's remark that MSRA50 and USPS prefer another γ is itself label-informed), and the dimension is the largest admissible grid value, not an accuracy pick.

The paper says both "repeat each clustering 50 times ... and report the results corresponding to the best objective function values" and prints mean±standard deviation. We do not invent a rule that turns one sentence into the other. Summaries, when generated, are mean±sample standard deviation over completed trials of the selected setting.

ACC is Equation (19) of Chang et al.: the fraction of points correct after the optimal one-to-one permutation of cluster ids [Munkres, 1957]. NMI is Equations (20)–(21), the Strehl–Ghosh geometric-mean form [Strehl and Ghosh, 2002], matching `average_method='geometric'` in scikit-learn [Pedregosa et al., 2011]. Labels are used only for these scores.

Software is Python 3.13.3 with NumPy 2.4.2, SciPy 1.17.1, and scikit-learn 1.8.0. Hardware recorded for this work is 8 CPU cores and 16 GB RAM. No GPU was used. Per-solve IRLS wall times are in Table 7. Code and run records are intended for `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` branch `v1`.

5 Experimental results

Chang et al. report, in their Tables 1 and 2, mean ACC and NMI with standard deviations on all eight named datasets, and they state that SSC is "consistently the best algorithm using both evaluation metrics." Those numbers are claims of Chang et al. (2015). They are labelled "Chang et al. (2015)" in our tables and are not Our Reproduction.

Our clustering statistics come only from `scripts/generate_report_tables.py` reading completed records under `runs/`. There are 191 run directories: 184 `COMPLETED`, 6 `SKIPPED` (all SPLS), and 1 `FAILED` (Track A LDA+K-means on ORL at dimension 20; a later job with the same setting completed). Summaries are mean±sample standard deviation over *completed* trials of the selected setting. We never pad missing trials to 50.

Track A (label-informed). Table 2 and Figure 1 compare Hungarian ACC on the four obtained datasets. AR, UMIST, BinAlpha, and MSRA50 are omitted because no local file exists; those cells would be "N/A — not reproduced." SPLS is "N/A — not reproduced" on every dataset. LDA-SC is approximate and is kept off this table.

The main negative finding is on ORL. Our SSC, after selecting the γ with the highest mean ACC, is 53.9 ± 3.0 ($n = 50$, $\gamma = 10^{-6}$). Chang et al. report 76.0 ± 1.6. Our spectral clustering baseline is 54.2 ± 3.1 ($n = 50$). Under this protocol SSC is statistically indistinguishable from SC and does *not* reproduce the paper's large ORL gain. K-means on the same Olivetti-derived matrix is 51.8 ± 3.3 ($n = 50$) versus the paper's 60.5 ± 1.8, so the gap is not confined to SSC.

JAFFE is the closest match. Track A SSC is 74.8 ± 8.4 ($n = 50$) versus 81.2 ± 1.6 in the paper. Label-informed selection chose $\gamma = 10^6$, not 1. At $\gamma = 1$ the same method is 68.6 ± 11.3 ($n = 50$). The paper's remark that $\gamma = 1$ is usually good does not hold on this JAFFE run. LLC on JAFFE *exceeds* the paper (88.4 ± 6.9, $n = 50$, versus 78.6 ± 1.5). That is a real run, not a copied cell, and it does not license replacing our SSC numbers by LLC.

YaleB SSC is 19.7 ± 1.1 with $n = 10$, not 50, versus the paper's 49.8 ± 1.6. USPS SSC is 62.1 with $n = 1$, not 50, versus 75.5 ± 1.9. Those rows are not a 50-run reproduction. PCA, LDA, LLC, and SC were not completed on USPS; PCA, LDA, and LLC were not completed on YaleB. NMI (Table 3) tells the same story: ORL SSC 76.5 ± 1.7 ($n = 50$) versus the paper's 88.6 ± 1.5.

Track B (no labels in training or tuning). Table 4 freezes $\gamma = 1$ and the prespecified dimension $\min(50, c, n - 1, d)$ (LDA: $c - 1$). On ORL, SSC is 53.7 ± 2.8 ($n = 50$), again next to SC at 54.2 ± 3.1. On JAFFE, SSC at the frozen $\gamma = 1$ is 68.6 ± 11.3 ($n = 50$), below Track A's selected $\gamma = 10^6$. USPS SSC is the single completed $\gamma = 1$ job (Track A directory), $n = 1$; it is marked with a dagger rather than invented Track B repeats. NMI is in Table 5.

γ sensitivity and IRLS. Table 6 and Figure 2 show the Track A SSC γ grid where it was actually run. On ORL, $\gamma \in \{10^{-6}, 10^{-3}, 1\}$ are essentially tied near 54% ACC ($n = 50$); $\gamma = 10^3$ and 10^6 collapse to 43.1 ± 2.6 and 36.1 ± 2.9. On JAFFE the opposite happens: ACC rises with γ, and 10^6

Table 2: Track A Hungarian ACC (%). "Chang et al. (2015)" quotes Tables 1–2 of the paper. "Our Reproduction" is $100\times$ mean$\pm$sample standard deviation over completed trials of the label-informed selected setting. Trial counts are actual: ORL/JAFFE SSC $n = 50$; YaleB SSC $n = 10$; USPS SSC $n = 1$. PCA-SC and LDA-KM used $n = 10$ on ORL/JAFFE. AR, UMIST, BinAlpha, and MSRA50 are unavailable. SPLS is not implemented. LDA-SC is omitted.

Method	Source	ORL	JAFFE	YaleB	USPS
K-means	Chang et al. (2015)	60.5±1.8	75.6±1.8	14.4±1.5	65.4±1.7
K-means	Our Reproduction	51.8±3.3 (n=50)	73.2±5.7 (n=50)	9.3±0.6 (n=10)	66.3±1.7 (n=10)
SC	Chang et al. (2015)	72.7±2.3	76.1±1.6	34.8±1.4	64.3±1.4
SC	Our Reproduction	54.2±3.1 (n=50)	67.7±11.3 (n=50)	20.1±1.2 (n=10)	N/A — not reproduced
PCA-KM	Chang et al. (2015)	64.5±2.3	75.8±1.5	24.9±1.8	69.4±1.8
PCA-KM	Our Reproduction	53.0±3.1 (n=50)	74.8±7.7 (n=50)	N/A — not reproduced	N/A — not reproduced
PCA-SC	Chang et al. (2015)	67.8±2.1	76.9±1.8	36.8±2.0	69.1±1.5
PCA-SC	Our Reproduction	54.9±2.7 (n=10)	69.8±8.3 (n=10)	N/A — not reproduced	N/A — not reproduced
LDA-KM	Chang et al. (2015)	65.6±2.6	76.5±1.7	26.1±1.8	70.1±1.3
LDA-KM	Our Reproduction	48.5±3.7 (n=10)	64.8±11.7 (n=10)	N/A — not reproduced	N/A — not reproduced
LLC	Chang et al. (2015)	71.5±2.2	78.6±1.5	38.2±1.5	63.9±1.7
LLC	Our Reproduction	62.1±2.0 (n=50)	88.4±6.9 (n=50)	N/A — not reproduced	N/A — not reproduced
SPLS	Chang et al. (2015)	74.2±1.8	79.5±2.1	47.3±1.7	71.4±1.6
SPLS	Our Reproduction	N/A — not reproduced	N/A — not reproduced	N/A — not reproduced	N/A — not reproduced
SSC	Chang et al. (2015)	76.0±1.6	81.2±1.6	49.8±1.6	75.5±1.9
SSC	Our Reproduction	53.9±3.0 (n=50)	74.8±8.4 (n=50)	19.7±1.1 (n=10)	62.1 (n=1)

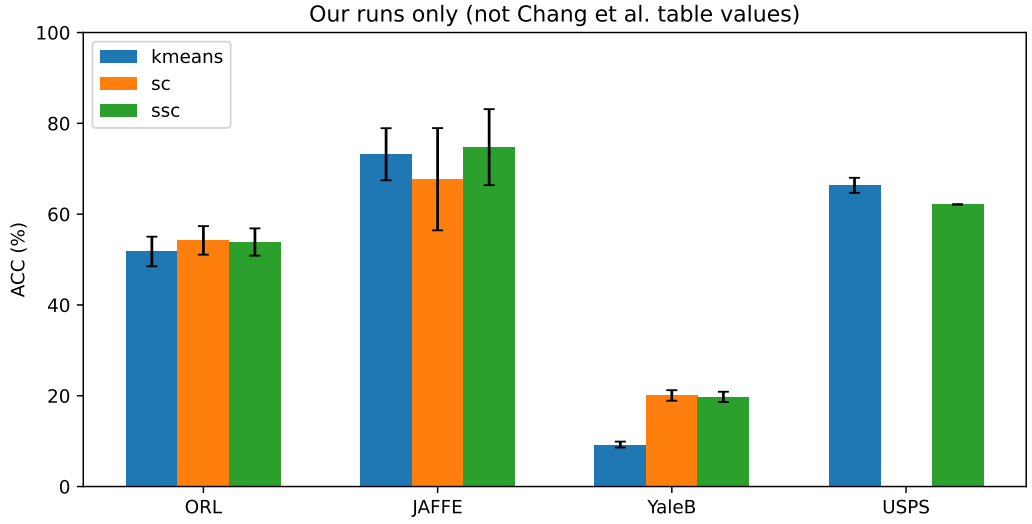


Figure 1: Track A Hungarian ACC (%) for K-means, spectral clustering, and SSC on obtained datasets. Bars are Our Reproduction means; error bars are sample standard deviations. ORL and JAFFE use $n = 50$; YaleB uses $n = 10$; USPS K-means uses $n = 10$ and SSC uses $n = 1$. SC was not run on USPS. These bars are not Chang et al. table entries.

Table 3: Track A NMI (%) with geometric-mean normalization. Same selection protocol and trial counts as Table 2.

Method	Source	ORL	JAFFE	YaleB	USPS
K-means	Chang et al. (2015)	80.3±1.8	79.4±0.8	17.3±1.5	67.3±1.8
K-means	Our Reproduction	74.4±1.5 (n=50)	83.1±3.5 (n=50)	13.0±0.9 (n=10)	61.2±0.4 (n=10)
SC	Chang et al. (2015)	85.8±1.9	80.2±0.9	55.6±1.6	69.5±1.6
SC	Our Reproduction	76.5±1.6 (n=50)	82.8±5.7 (n=50)	29.9±0.9 (n=10)	N/A — not reproduced
PCA-KM	Chang et al. (2015)	80.6±1.6	79.8±0.8	26.7±1.8	73.1±1.9
PCA-KM	Our Reproduction	75.2±1.4 (n=50)	83.1±4.5 (n=50)	N/A — not reproduced	N/A — not reproduced
PCA-SC	Chang et al. (2015)	86.3±1.4	81.5±1.3	38.6±1.5	74.2±1.8
PCA-SC	Our Reproduction	77.3±1.1 (n=10)	84.0±4.4 (n=10)	N/A — not reproduced	N/A — not reproduced
LDA-KM	Chang et al. (2015)	81.1±2.1	82.1±1.4	29.4±1.6	75.1±1.6
LDA-KM	Our Reproduction	72.5±2.1 (n=10)	78.7±6.0 (n=10)	N/A — not reproduced	N/A — not reproduced
LLC	Chang et al. (2015)	84.9±1.5	82.5±1.7	34.1±1.3	67.5±1.5
LLC	Our Reproduction	78.7±1.3 (n=50)	91.2±2.7 (n=50)	N/A — not reproduced	N/A — not reproduced
SPLS	Chang et al. (2015)	87.2±1.8	83.1±2.1	41.4±1.6	76.5±1.8
SPLS	Our Reproduction	N/A — not reproduced	N/A — not reproduced	N/A — not reproduced	N/A — not reproduced
SSC	Chang et al. (2015)	88.6±1.5	84.3±1.6	46.8±1.3	79.8±1.6
SSC	Our Reproduction	76.5±1.7 (n=50)	85.6±4.3 (n=50)	29.5±1.0 (n=10)	73.3 (n=1)

Table 4: Track B Hungarian ACC (%). No labels for training or tuning; $\gamma = 1$ and the pre-fixed dimension rule. Mean±sample standard deviation over completed trials. YaleB uses $n = 10$. †USPS SSC is the one completed $\gamma = 1$ run ($n = 1$), reused from the Track A job; no separate Track B USPS SSC job was launched.

Method	ORL	JAFFE	YaleB	USPS
K-means	51.8±3.3 (n=50)	73.2±5.7 (n=50)	9.3±0.6 (n=10)	N/A — not reproduced
SC	54.2±3.1 (n=50)	67.7±11.3 (n=50)	20.1±1.2 (n=10)	N/A — not reproduced
PCA-KM	52.6±2.9 (n=50)	74.8±7.7 (n=50)	9.1±0.6 (n=10)	N/A — not reproduced
PCA-SC	53.0±3.1 (n=50)	60.0±7.4 (n=50)	N/A — not reproduced	N/A — not reproduced
LDA-KM	48.5±3.0 (n=50)	64.4±9.2 (n=50)	N/A — not reproduced	N/A — not reproduced
LLC	62.1±2.0 (n=50)	88.4±6.9 (n=50)	N/A — not reproduced	N/A — not reproduced
SSC	53.7±2.8 (n=50)	68.6±11.3 (n=50)	19.7±1.1 (n=10)	62.1 (n=1)†

Table 5: Track B NMI (%). Same protocol and trial counts as Table 4.

Method	ORL	JAFFE	YaleB	USPS
K-means	74.4±1.5 (n=50)	83.1±3.5 (n=50)	13.0±0.9 (n=10)	N/A — not reproduced
SC	76.5±1.6 (n=50)	82.8±5.7 (n=50)	29.9±0.9 (n=10)	N/A — not reproduced
PCA-KM	75.0±1.5 (n=50)	83.1±4.5 (n=50)	12.7±0.9 (n=10)	N/A — not reproduced
PCA-SC	76.1±1.8 (n=50)	73.2±4.0 (n=50)	N/A — not reproduced	N/A — not reproduced
LDA-KM	72.9±1.7 (n=50)	78.6±5.1 (n=50)	N/A — not reproduced	N/A — not reproduced
LLC	78.7±1.3 (n=50)	91.2±2.7 (n=50)	N/A — not reproduced	N/A — not reproduced
SSC	76.7±1.5 (n=50)	83.0±5.8 (n=50)	29.5±1.0 (n=10)	73.3 (n=1)†

is best. YaleB and USPS have only $\gamma = 1$. Table 7 records IRLS at $\gamma = 1$: ORL and YaleB hit `max_iter`$= 300$ without meeting `tol`$= 10^{-6}$; JAFFE converged in 263 iterations; the single USPS solve converged in 171 iterations (7.89 s).

Table 6: Track A SSC γ grid on obtained datasets. ACC and NMI are $100\times$ mean$\pm$sample standard deviation over completed trials. ORL and JAFFE: $n = 50$ at each γ. YaleB: $\gamma = 1$ only, $n = 10$. USPS: $\gamma = 1$ only, $n = 1$.

Dataset	γ	ACC%	NMI%	n
ORL	10^{-6}	53.9±3.0 (n=50)	76.5±1.7 (n=50)	50
ORL	10^{-3}	53.9±3.0 (n=50)	76.5±1.7 (n=50)	50
ORL	1	53.7±2.8 (n=50)	76.7±1.5 (n=50)	50
ORL	10^{3}	43.1±2.6 (n=50)	66.0±1.3 (n=50)	50
ORL	10^{6}	36.1±2.9 (n=50)	57.0±1.3 (n=50)	50
JAFFE	10^{-6}	68.5±11.2 (n=50)	83.0±5.7 (n=50)	50
JAFFE	10^{-3}	68.5±11.2 (n=50)	83.0±5.7 (n=50)	50
JAFFE	1	68.6±11.3 (n=50)	83.0±5.8 (n=50)	50
JAFFE	10^{3}	70.6±9.7 (n=50)	83.6±5.1 (n=50)	50
JAFFE	10^{6}	74.8±8.4 (n=50)	85.6±4.3 (n=50)	50
YaleB	1	19.7±1.1 (n=10)	29.5±1.0 (n=10)	10
USPS	1	62.1 (n=1)	73.3 (n=1)	1

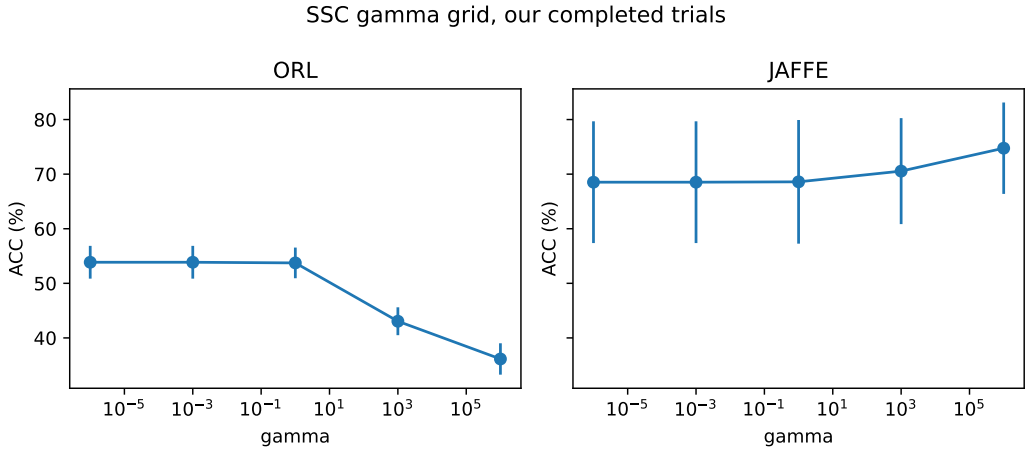


Figure 2: Track A SSC ACC (%) versus γ on ORL and JAFFE. Each point is the mean over $n = 50$ completed K-means trials; error bars are sample standard deviations. YaleB and USPS are omitted because only $\gamma = 1$ was run.

6 Method extension

Observed limitation: SSC's shrinkage graph W includes a union-kNN edge whenever either embedded point lists the other, so a single noisy neighbour can contribute a fusion pull. Research hypothesis: replacing that support by a mutual (AND) kNN graph on F, with the same nonnegative Gaussian weights frozen before IRLS, should remove some one-sided edges without leaving the convexity setting of Equation (2).

Mutual kNN is an established graph [Brito et al., 1997, von Luxburg, 2007]. Maier et al. [2009] already warn that the AND rule can disconnect clusters, especially at small k; SSC's experimental $k = 5$ is therefore a risk, not a new rate. Chi and Lange [2015] freeze sparse Gaussian weights for convex clustering, but their indicator is union (OR) kNN, not mutual, and their variables are centroids of the original points. Tan et al. [2020] use mutual kNN inside ordinary spectral clustering, i.e. on the affinity that defines the embedding, not on an SSC shrinkage step after F is already computed. We did not find a paper that applies mutual kNN to SSC's W in Equation (2). That absence justifies calling the change a new combination, not a new theorem: nonnegative frozen W is exactly the hypothesis Chang et al. use for convexity.

Table 7: IRLS at $\gamma = 1$: iteration count, whether the relative Frobenius step met 10^{-6} before `max_iter`$= 300$, final unsmoothed Equation (2), trial count, and wall time in seconds. SSC rows are Track A; RSSC rows are EX1. USPS SSC is $n = 1$. The unsmoothed objective is not declared equal to the square-root-smoothed surrogate.

Dataset	Method	γ	IRLS iter.	Converged	Eq. (8)	n	sec.
ORL	SSC	1	300	no	29.2	50	0.72
ORL	RSSC	1	300	no	24.2	50	0.46
JAFFE	SSC	1	263	yes	1.97	50	0.30
JAFFE	RSSC	1	298	yes	1.67	50	0.28
YaleB	SSC	1	300	no	47.7	10	6.71
YaleB	RSSC	1	300	no	30.6	10	2.60
USPS	SSC	1	171	yes	14.8	1	7.89

Implementation (in `method_extension/`, separate from `SSC_method/`): after computing F as in Section 3, keep an edge only when each point is among the other's k neighbours, place the usual Gaussian weight, and pass that W into the existing IRLS solver. Weights are not updated from labels and are not moved inside the iteration.

Table 8 and Figure 3 compare original SSC (A0) with this mutual-kNN variant (A1, RSSC) at $\gamma = 1$, $k = 5$. On ORL, A0 is 53.7 ± 2.8 and A1 is 54.0 ± 2.8 ($n = 50$). On JAFFE, 68.6 ± 11.3 versus 68.7 ± 11.3 ($n = 50$). On YaleB, 19.7 ± 1.1 versus 19.9 ± 1.3 ($n = 10$). Every difference sits inside the trial standard deviation. USPS RSSC was not run; the SSC cell is the $n = 1$ $\gamma = 1$ job. The outcome is mixed or negligible, not a win, and we do not claim an improvement in ACC, NMI, robustness, or runtime.

Table 8: Original SSC (A0, union kNN W) versus mutual-kNN W on F (A1, RSSC) at $\gamma = 1$, $k = 5$. ACC and NMI are $100\times$ mean$\pm$sample standard deviation. ORL and JAFFE: $n = 50$. YaleB: $n = 10$. USPS: SSC $n = 1$, RSSC not reproduced.

Dataset	A0 SSC ACC	A1 RSSC ACC	A0 SSC NMI	A1 RSSC NMI	n
ORL	53.7±2.8	54.0±2.8	76.7±1.5	76.6±1.5	50
JAFFE	68.6±11.3	68.7±11.3	83.0±5.8	83.0±5.9	50
YaleB	19.7±1.1	19.9±1.3	29.5±1.0	29.8±1.2	10
USPS	62.1±0.0	N/A — not reproduced	73.3±0.0	N/A — not reproduced	1

7 Ablation and robustness

The ablations are A0, original union-kNN SSC, versus A1, mutual-kNN W on F with IRLS unchanged. Table 9 and Figure 4 vary $k \in \{3, 5, 7, 9\}$ on ORL and JAFFE ($n = 20$ trials per setting) and include YaleB only at $k = 5$ ($n = 10$). On ORL the two methods stay within about one point of each other at every k; A1 is slightly higher at $k = 7$ (53.0 ± 3.0 versus 51.7 ± 2.4) but the intervals overlap. On JAFFE, A1 never pulls away from A0. Mutual kNN is not the component that would close the gap to the 2015 tables.

Table 10 summarizes EX2–EX5. Each cell is the unweighted mean of per-seed mean ACC, with the standard deviation of those seed-level means; each seed used $n = 20$ K-means trials. EX2 (Gaussian noise at 0.1 and 0.3 times $\mathrm{std}(X)$) and EX5 (drop 10% of shrinkage-graph edges; two seeds, not three) leave A0 and A1 tied. EX4 (replace 5% of samples by $\mathcal{N}(0, 10)$) is the same. EX3 (replace 10% of features by $\mathcal{N}(0, 1)$) hurts both methods: ORL SSC falls to 18.2 ± 0.6 versus A1 19.8 ± 0.3 (three seeds, $n = 20$ each). That is a small A1 edge after a large collapse, not a robustness win.

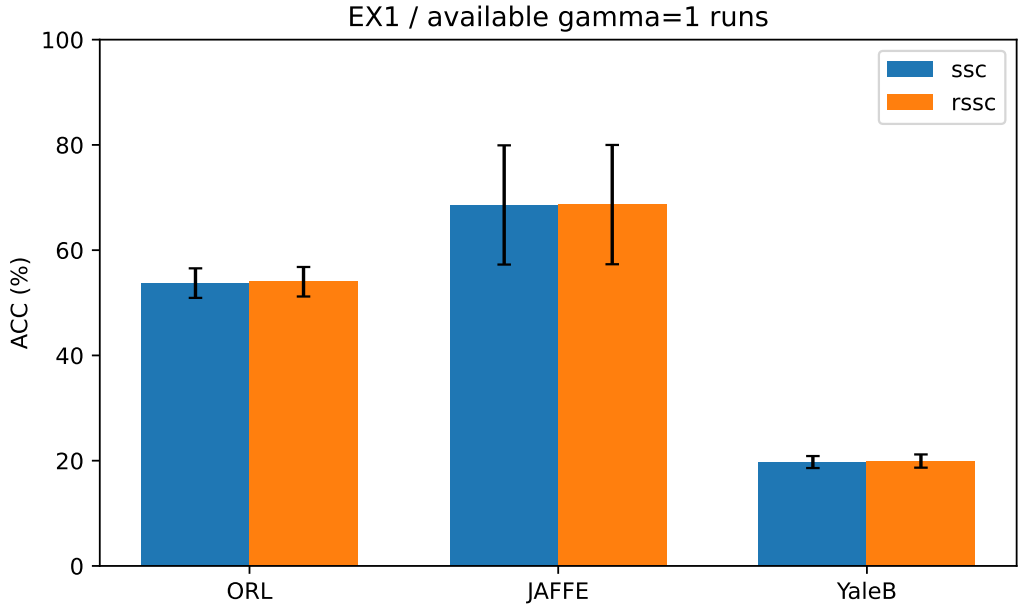


Figure 3: EX1 ACC (%) for original SSC versus mutual-kNN RSSC on ORL ($n = 50$), JAFFE ($n = 50$), and YaleB ($n = 10$), all at $\gamma = 1$, $k = 5$. Error bars are sample standard deviations. USPS RSSC was not run.

Negative results are kept. These stresses were not run on YaleB or USPS and were not repeated 50 times.

Table 9: Ablation A0 (union kNN SSC) versus A1 (mutual kNN only) while varying neighbourhood size. ACC and NMI are 100× mean±sample standard deviation. ORL and JAFFE: $n = 20$ at each k. YaleB: $k = 5$ only, $n = 10$.

Dataset	k	A0 SSC ACC	A1 RSSC ACC	A0 NMI	A1 NMI	n
ORL	3	52.8±3.9	52.5±4.2	76.6±1.6	76.5±1.8	20
ORL	5	53.8±2.9	54.1±2.6	76.6±1.5	76.5±1.3	20
ORL	7	51.7±2.4	53.0±3.0	74.9±1.8	75.7±1.6	20
ORL	9	51.6±2.8	52.6±2.7	74.7±1.8	74.9±1.2	20
JAFFE	3	68.1±9.6	68.2±9.8	82.6±4.7	82.6±4.7	20
JAFFE	5	67.8±11.3	68.6±11.3	82.8±5.8	83.0±5.8	20
JAFFE	7	67.7±9.1	68.1±9.0	83.0±4.9	83.2±5.2	20
JAFFE	9	67.2±9.1	67.8±8.9	82.3±4.7	82.4±4.8	20
YaleB	5	19.7±1.1	19.9±1.3	29.5±1.0	29.8±1.2	10

8 Critical discussion

SSC's main modelling claim is specific: the graph that should regularize the embedding is a graph on the embedding, not the graph that produced the embedding. That claim is independent of the later typesetting errors, and it is a reasonable response to the COMP3314 observation that a neighbourhood in pixel space need not be a neighbourhood after a nonlinear map.

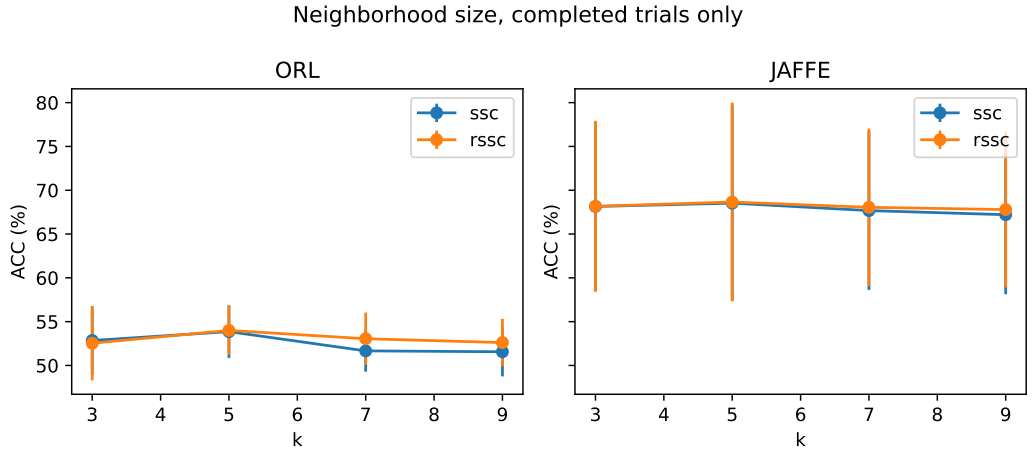


Figure 4: ACC (%) versus neighbourhood size k for A0 SSC and A1 RSSC on ORL and JAFFE. Each point uses $n = 20$ completed trials; error bars are sample standard deviations.

Table 10: Robustness stresses on ORL and JAFFE. EX2: isotropic Gaussian noise with $\sigma \in \{0.1, 0.3\} \times \mathrm{std}(X)$. EX3: 10% of features replaced by $\mathcal{N}(0, 1)$. EX4: 5% of samples replaced by $\mathcal{N}(0, 10)$. EX5: drop 10% of shrinkage-graph edges. Reported ACC is the mean of per-seed mean ACC $\pm$ the standard deviation of those seed-level means; each seed used $n = 20$ trials. EX2–EX4 used three perturbation seeds; EX5 used two. Not a 50-run protocol.

Exp.	Dataset	Level	SSC ACC	RSSC ACC	seeds	n/seed
EX2	JAFFE	0.1	67.1±1.1 (n=20)	66.9±1.2 (n=20)	3	20
EX2	JAFFE	0.3	68.9±1.3 (n=20)	68.7±1.2 (n=20)	3	20
EX2	ORL	0.1	53.3±1.0 (n=20)	53.4±0.9 (n=20)	3	20
EX2	ORL	0.3	53.4±0.2 (n=20)	53.4±0.8 (n=20)	3	20
EX3	JAFFE	0.1	39.5±1.1 (n=20)	41.1±0.7 (n=20)	3	20
EX3	ORL	0.1	18.2±0.6 (n=20)	19.8±0.3 (n=20)	3	20
EX4	JAFFE	0.05	64.1±1.7 (n=20)	64.4±1.5 (n=20)	3	20
EX4	ORL	0.05	50.8±0.8 (n=20)	50.9±0.3 (n=20)	3	20
EX5	JAFFE	0.1	67.8±0.1 (n=20)	67.9±0.1 (n=20)	2	20
EX5	ORL	0.1	53.8±0.6 (n=20)	54.4±0.2 (n=20)	2	20

The published optimization story is weaker than the modelling claim. Equation (10), Algorithm 1 line 10, and the factor-of-two gap between Equation (2) and Equation (4) are not minor typos in isolation: they affect the matrix that is inverted, the right-hand side, and the meaning of γ. A reader who implemented line 10 literally would not be solving SSC. Because no official code was available, those ambiguities are not resolvable by "run the authors' script." Our locked choices make the solver internally consistent with Equation (4), at the cost of no longer matching printed Equation (2) as a numerical instance.

The experimental section of Chang et al. is also less decisive than the abstract suggests. "Report the best results" together with a γ grid and a subspace range of 10 to 150 is a label-informed search. Mean±std next to "best objective" is unresolved. Four of eight datasets are not currently obtainable from a public file that matches the stated counts, so even a perfect solver cannot reproduce the full tables. SPLS, described as the second-best method, is not implementable from the open abstract. LDA+SC is under-specified. Under those conditions, copying Table 1 of Chang et al. into a reproduction paper would not be evidence.

The mutual-kNN adaptation is modest on purpose. Moving W inside the IRLS loop, or replacing W by the adaptive neighbours of Nie et al. [2014], would require a separate convexity argument. We did not take that step. Mutual kNN can help by deleting bridges; it can also fragment a cluster [Maier et al., 2009]. On the obtained images the measured difference is mixed or negligible (Table 8), so the adaptation does not repair the ORL or YaleB gap versus Chang et al.

Relative to 2015, the method remains a small convex post-process on a classical spectral embedding. It does not use deep features, and it still ends with K-means. Its present value, for this course, is as a worked example of graph construction, eigenmaps, robust losses, and the difference between a printed algorithm and a well-defined convex program.

9 Limitations

The following limitations are in force in this PDF.

Missing datasets: AR, UMIST, BinAlpha, and MSRA50 were not obtained; paper counts for those names were not measured on a local file. ORL is a 64×64 Olivetti variant, not 92×112. YaleB is a pre-resized mat whose interpolation is unknown.

Baselines: SPLS is not implemented. LDA+spectral clustering is an approximation. LLC uses the Wu–Schölkopf equations with SSC's $k = 5$, not LLC's original search grid.

Unspecified quantities: δ, G_0, the K-means `n_init`, and the step of the 10–150 dimension grid are not in the paper. Median bandwidth, $G_0 = F$, and dimension step 10 are assumptions. The 50-run versus best-objective sentence is left unresolved.

Coefficient: the solver implements γ in Equation (4), equivalent to $\gamma/2$ in Equation (2).

No original code: numerical identity with the 2015 implementation cannot be checked.

Experiments: YaleB used 10 trials, not 50; USPS SSC used 1 trial, not 50. USPS is missing SC, PCA, LDA, LLC, and RSSC. YaleB Track A is missing PCA, LDA, and LLC. The γ grid was not run on YaleB or USPS. IRLS at $\gamma = 1$ on ORL and YaleB did not meet the 10^{-6} step tolerance within 300 iterations. ORL is an Olivetti 64×64 resize; YaleB is a pre-resized mat. Those facts bound any claim that we reproduced Tables 1–2 of Chang et al.

Face images: ORL, JAFFE, and YaleB are identity-labelled faces collected by others. This project does not collect new human-subjects data and does not release the images.

10 Conclusion

We reimplemented SSC from Chang et al. as a normalized-cut embedding, a frozen kNN fusion weight, and IRLS for Equation (4), with Equation (10) and Algorithm 1 line 10 treated as typesetting errors. Four datasets are in hand; four are not. SPLS is not implemented. On the obtained data, SSC did not reproduce the paper's large ORL gain (53.9 ± 3.0, $n = 50$, versus 76.0), and it is indistinguishable from our spectral clustering baseline. YaleB used 10 trials and USPS SSC used 1 trial. A mutual-kNN affinity on F is an adaptation of a standard graph to SSC's already convex shrinkage step; empirically it is mixed or negligible, not a win. The honest summary is the protocol, the corrections, the missing data, and those negative findings—not a claim that the 2015 ranking has been reproduced.

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This work is an HKU COMP3314 course report. No external funding is claimed. Group membership, names, and emails require confirmation and are not invented here. Code is intended for `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` on branch `v1`.

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A Equation audit

This appendix records the locked implementation relative to Chang et al., without repeating benchmark statistics.

Laplacian identity. For a symmetric nonnegative W and combinatorial $\widetilde{L} = \widetilde{D} - \widetilde{W}$,

$$\sum_{ij} \widetilde{W}_{ij} \|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L} G).$$

A hand example with $G = (0.2, 0.8)^\top$, $W_{01} = W_{10} = 1$, and distance 0.6 gives $\sum_{ij} W_{ij}\|g_i - g_j\|_2^2 = 0.72$, $\sum_{ij} W_{ij}\|g_i - g_j\|_2 = 1.2$, $\widetilde{W}_{01} = 5/6$, $\sum_{ij} \widetilde{W}_{ij}\|g_i - g_j\|_2^2 = 0.6$, and $\mathrm{Tr}(G^\top \widetilde{L} G) = 0.3$. Hence, at the expansion point,

$$\sum_{ij} \widetilde{W}_{ij} \|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L} G) = \tfrac{1}{2} \sum_{ij} W_{ij} \|g_i - g_j\|_2.$$

Gradient of Equation (2)**.** For symmetric W,

$$\frac{\partial}{\partial g_k} \sum_{ij} W_{ij} \|g_i - g_j\|_2 = 4(\widetilde{L}G)_k, \qquad \frac{\partial}{\partial g_k} \|h_k\|_2 = 2S_{kk}(g_k - f_k).$$

Setting the gradient to zero yields $(S + 2\gamma\widetilde{L})G = SF$. Printed Equation (13) omits the 2.

Independent convex check. On a synthetic chain with $n = 6$, $c = 2$, $\gamma = 0.8$, and random seed 1, a unit test compared the IRLS output of Equation (4) to CVXPY/CLARABEL solutions of the two convex problems. The G from Equation (4) matched the $\gamma/2$ instance of Equation (2) (objective about 3.844, largest absolute entry difference about 2×10^{-5}) and did not match the minimizer of printed Equation (2) (printed objective about 4.537 at that G; printed optimum about 4.021). Those figures are solver-audit values on a synthetic graph, not clustering scores, and they are not copied from Chang et al.

What is monitored. `equation8_objective` is unsmoothed Equation (2). `sqrt_smoothed_equation8_objective` replaces each Euclidean norm by $\sqrt{\|v\|_2^2 + \varepsilon^2}$. The IRLS denominator floor $\max(\|v\|_2, \varepsilon)$ is a third function. Linear residuals are $\|(S + \gamma\widetilde{L})G - SF\|_F$ with $S, \widetilde{L}$ frozen during an iteration; `fixed_point_residual` recomputes S and $\widetilde{L}$ at the final G.

B Dataset manifests

Checksums and URLs below were recorded on 2026-10-20. Raw images are gitignored.

ORL (approximate). Source `https://ndownloader.figshare.com/files/5976027`. The SHA-256 is stored in `benchmarks/orl/manifest.yaml` and matches scikit-learn 1.8.0 Olivetti. Credit AT&T Laboratories Cambridge [Samaria and Harter, 1994]. Layout: transpose and reshape to $(400, 64, 64)$, bilinear 32×32, divide by 255.

JAFFE (verified). Zenodo record 14974867 [Lyons et al., 2025]; SHA-256 in `benchmarks/jaffe/manifest.yaml`. Non-commercial research only; redistribution forbidden; at most ten images in a publication. 213 TIFFs, all 256×256; 23 palette-mode files converted after checking $R = G = B$. Subject codes KA, KL, KM, KR, MK, NA, NM, TM, UY, YM.

YaleB (approximate). Deng Cai file `YaleB_32x32.mat` from the CAD ZJU FaceData pages; SHA-256 in `benchmarks/yaleb/manifest.yaml`. Research use; cite Georghiades et al. [2001] and the cropped protocol of Lee et al. [2005]. UCSD cropped zips returned 404. No loader-side resize.

USPS (verified). LIBSVM `usps.bz2` and `usps.t.bz2`; both SHA-256 values are in `benchmarks/usps/manifest.yaml`. $7291 + 2007$ rows, 256 features, labels mapped from 1..10 to 0..9. No licence file inside the archives.

Unavailable. AR: Ohio State 503, Purdue 404; NIST catalog is not the image archive [Martinez and Benavente, 1998]. UMIST: Lincoln eprints HTTP 202 with WAF HTML; Sheffield URL 404 [Graham and Allinson, 1998]. BinAlpha: no traceable 26×30 file; Binary Alphadigits not substituted. MSRA50: no measured 1799×12 archive; unofficial CSDN mats unused.

C Protocol assumptions

Explicit in Chang et al.: $k = 5$ for spectral methods; γ grid $\{10^{-6}, 10^{-3}, 1, 10^3, 10^6\}$; 50 repetitions with random initialization; ACC with Kuhn–Munkres; NMI with geometric-mean entropy; subspace "10 to 150" for dimensionality-reduction baselines; "report the best results."

Unspecified: bandwidth δ; matrix G_0; K-means `n_init`; dimension-grid step.

Assumptions adopted in `original_paper.yaml`: δ^2 is the median positive kNN squared distance, separately on the feature graph and on the embedding graph; $G_0 = F$; dimension grid $\{10, 20, \ldots, 150\}$ clipped by n and d. The protocol file does not adopt a numeric `n_init`. Track B freezes $\gamma = 1$ and the largest admissible grid dimension without reading labels.

Ambiguity left open: the conjunction of "best objective function values" with printed mean$\pm$standard deviation.

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